



Systems Engineering Documentation: Warehouse Manager

Role Tier: Level 2 (Regional Administrator)

Primary Objective: To manage the intake, storage, accuracy, and outbound processing of both **physical goods** and **human resources** (Labor/Packing teams) within an assigned hub.

1 Inventory & Stock Management

The WM governs the state of goods before they become orders.

- **Real-Time SKU Tracking:** A live dashboard displaying SKU ID, quantity, dimensions, and storage location (Aisle/Shelf/Bin).
- **Safety Stock Alerts:** Automated alerts when inventory drops below thresholds, with escalation to the Logistics Manager.
- **Inbound Receiving:** Scanning and verifying vendor shipments against Advance Shipping Notices (ASN) and flagging discrepancies immediately.

2 Service & Resource Management (Individual Moves)

Since your system supports individual "House Shifts," the WM now manages **Field Resources**.

- **Labor Roster & Deployment:** A live status tracker of laborers (Status: *In-Hub*, *On-Field*, *Off-Duty*). The WM assigns specific "Helpers" to a Driver's truck based on the customer's request.
- **Utensil & Moving Supply Inventory:** Dedicated tracking for "Service Items" like plastic crates, padded blankets, and wardrobe boxes.
- **Returnable Asset Tracking:** Monitors equipment that goes out to a customer's house and ensures it is scanned back into the warehouse after the move.

3 Internal Mapping & Spatial Logic

The WM digitizes the physical floor to maximize picking speed and accuracy.

- **Location-Level Tracking:** Every SKU/Item is mapped to a specific Zone-Aisle-Rack-Shelf-Bin.
- **Digital Floor Plan:** A 2D grid representing the warehouse. Green bins are available; Red bins are occupied.
- **Dynamic Slotting:** Fast-moving items are placed near packing zones, and heavy items are kept on lower racks for safety.

4 Fulfillment & "The Picking Queue"

This process transforms static inventory into a mobile manifest.

- **Intelligent Pick-Lists:** The system generates paths that prevent "double-back" walking, saving time.
- **Packing Station Interface:** Verification scan of items, weight checks, and automatic shipping label/manifest generation.
- **Order Status Flow:** Orders transition from *Pending* → *Picked* → *Packed* → *Ready for Dispatch*.

5 Dispatcher & Driver Hand-Off (Loading Dock)

The WM controls the physical "Exit" node of the system.

- **Bay & Dock Management:** Assigning trucks to specific docks to prevent congestion.
- **Crew & Manifest Verification:** Ensuring the Driver has scanned all packages **AND** that the assigned **Laborers** are physically present in the truck.
- **Dwell Time Logging:** Automatically tracking the time from "Truck Arrival" to "Truck Loaded" to measure hub efficiency.

6 Reverse Logistics (Returns Processing)

- **Inspection Station:** Returned or rejected goods are scanned and photographed upon arrival.
- **Disposition Workflow:** The WM decides if an item is *Restocked*, *Sent to Claims (Damaged)*, or *Discarded*.

7 Simple Generative AI Integration

- **Predictive Labor Scaling:** AI predicts upcoming volume and suggests: "*High move volume tomorrow; suggest moving 3 staff from Receiving to the Packing Team.*"
- **Natural Language Query:** The WM can type, "*Which aisle has enough space for 10 new pallets?*" or "*List all workers currently on-field.*"
- **Visual Damage Assessment:** AI analyzes photos of returns to suggest if an item can be resold.

Key Data Interfaces

Interface	Data Exchanged
WM ↔ Vendor	Receiving confirmations and ASN reconciliation.
WM ↔ Individual	Labor counts, packing utensils, and "House Shift" requirements.

WM ↔ Dispatcher	"Ready-for-Pickup" signals and confirmed crew lists.
WM ↔ Manager	Inventory valuation, labor hours (payroll), and dwell time metrics.
